

ANKIT KHUSHWAHA

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Education

Indian Institute Of Technology Dharwad

Bachelor of Science-Master of Science, Physics

Dec. 2022 – Present

Dharwad, Karnataka

Technical Skills

Languages: C, C++, Python

Systems & Tools: Linux, QEMU, U-boot, GDB, GNU Make, Git, Vim

Core Skills: Linux Device Driver, Kernel Modules, Kernel debugging, Memory Management, System Programming, CLI

Kernel Subsystems: Character Drivers, Sysfs, Procfs, Timers, deferred work, Interrupt Handling

Familiar With: Shell Scripting, Computer Architecture, Web Development

Experience

Linux Foundation Mentorship (LFX '25) - Linux Kernel

Sept 2025 - Dec 2025

Mentee

Upstream Linux Kernel Patches

- Contributed 10+ patches across Linux kernel selftests in net, mm, tls, mptcp, and user_events subsystems.
- Analyzed syzbot bugs using QEMU and analyzed execution paths with ftrace & gdb to identify root causes.
- Performed memory error analysis using KASAN to detect and fix uninitialized variables and invalid memory usage.
- Collaborated with mentors and upstream contributors through mailing-list discussions, code reviews, & patch revisions.

Acadia Ventures

July 2025 - Aug 2025

Research and Development Intern

Remote

- Built a structured testing framework with fixtures to manage inter-test dependencies, streamlining test maintenance and enabling faster debugging cycles.
- Implemented secure JWT-based authentication with role-based access, strengthening API security and control.
- Built notification system using event triggers and async tasks, improving user engagement with timely alerts.

Open-Source Contribution

Scientific Python Libraries

SunPy - Stingray

- Contributed to **SunPy** and **Stingray**, open-source Python libraries for scientific data analysis.
- Enhanced time-series analysis, data processing, ingestion, and visualization capabilities across both libraries.
- Redesigned documentation UI and Improved data handling, utility functions, documentation, and maintainability.
- Contributed to CI/CD automation, release workflows, code reviews, and issue discussions with project maintainers.

Projects

Linux Device Driver - Various Kernel Subsystems



C, Linux Kernel, Device Driver

- Developed Linux kernel modules across multiple kernel subsystems targeting **Linux kernel v7.1.0**.
- Implemented and explored kernel mechanisms including **timers, workqueues, sysfs, procfs, input handling, and synchronization**.
- Tested and debugged kernel modules using **QEMU** and kernel debugging tools, validating behavior in an isolated environment.
- Implemented kernel-user space interfaces with **error handling and resource management** for reliable driver operation.
- Documented subsystem internals and implementation details through technical blogs at ankitkdev.com/blog.

CPU SchedulerSimulation



C, pthreads, Makefile, gdb, Debug Logging

- Implemented a multithreaded CPU scheduling simulator with separate arrival, scheduler, and wakeup threads to model process execution.
- Added support for FCFS, SJF, SRTF, and RR scheduling, analyzing turnaround, waiting, and completion times.
- Built a debug logging system for testing and configured clean production builds with logs disabled, ensuring maintainability.

Certifications

The Linux Foundation

Issued Aug 2025

LFD103: A Beginner's Guide to Linux Kernel Development

Credly

Udemy

Issued July 2026

LDD1: Linux Device Driver Programming With Beaglebone Black

Udemy